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RESEARCH PROJECT

Intelligent Tutor Based on Multi-Agent Architecture and Language Models for the Adaptive Learning of Ancash Quechua

I. PROBLEM STATEMENT

1.1 Problem Situation

Ancash Quechua represents one of the most important and complex branches of the Quechua language family in Peru. However, it faces a gradual process of linguistic displacement and digital vulnerability due to the dominance of Spanish in formal educational settings and interactive technology platforms. Despite Intercultural Bilingual Education (IBE) initiatives, urban and peri-urban students have access to few adaptive, interactive learning environments geared toward autonomous conversational practice in Ancash Quechua.

From the perspective of Computer Science and Artificial Intelligence (AI), Intelligent Tutoring Systems (ITS) and modern conversational assistants based on Large Language Models (LLMs) have proven highly effective in teaching languages with abundant digital resources (English, Spanish, Mandarin). However, Ancash Quechua is considered a low-digital-resource language, characterized by rich agglutinative morphology and complex suffixation; this causes severe failures in generic commercial language models attempting to process its grammar, resulting in hallucinations or erroneous translations.

Consequently, there is a critical need to design an Intelligent Tutoring System with a multi-agent architecture that pedagogically breaks down the teaching process (grammar agent, dialogue/conversation agent, evaluator agent, and cultural feedback agent) to offer a personalized, rigorous conversational experience tailored to the student's proficiency level.

1.2 Research Objectives

1.2.1 General Objective

To design and implement an intelligent tutor based on a multi-agent architecture and adaptive language models to enhance the learning of Ancash Quechua among university and continuing education students.

1.2.2 Specific Objectives

1. To design the multi-agent architecture (Pedagogical Agent, Morphophonological Linguistic Agent, Evaluator Agent, and Conversational Agent) for personalized learning.
2. To build and integrate a verified Ancash Quechua linguistic corpus using vector databases and RAG to ensure grammatically accurate responses.
3. To develop a prototype interactive conversational assistant (web/mobile interface) featuring immediate, interactive feedback capabilities.
4. To evaluate the intelligent tutor's effectiveness in terms of linguistic accuracy, system usability (SUS), and learning gains among a pilot group of students.

II. THEORETICAL FRAMEWORK

2.1 Background of the Problem

Internationally, Intelligent Tutoring Systems (ITS) have evolved toward multi-agent architectures geared toward language learning. Zheng et al. (2023) developed a conversational agent framework for second-language teaching, demonstrating that role separation (grammatical correction vs. dialogic fluency) increases student retention by 28% compared to monolithic chatbots.

At the national level, research such as that by Cusihamán and Huamán (2022) explored computational tools for the morphological analysis of Cusco-Collao Quechua, revealing that rule-based analyzers need to be complemented by probabilistic models due to the high productivity of Quechua suffixation. Likewise, recent work at UNMSM (Gonzales and Castro-Mamani, 2014) on machine translation for Ancashino Quechua highlights the scarcity of annotated parallel data as the primary limitation, recommending the use of RAG techniques and few-shot prompting with

adapted foundation models.

2.2 Theoretical Foundations

- a) Intelligent Tutoring Systems (ITS) and Multi-Agent Systems (MAS): An ITS adapts instruction to the student's pace by structuring four traditional modules: the Student Model, Domain Model, Pedagogical Model, and Interface Model. In this project, each module is handled by a specialized agent with autonomous reasoning capabilities.
- b) Natural Language Processing for Agglutinative Languages (Low-Resource NLP): Ancashino Quechua employs derivation and inflection through chained suffixes. Agglutinative morphological theory guides tokenization and response validation to prevent the erroneous fragmentation of complex terms.
- c) RAG (Retrieval-Augmented Generation) Architecture: A technique that combines the generative capability of LLMs with semantic search within an external vector knowledge base (Ancash dictionaries and grammars), ensuring linguistic accuracy and reducing hallucinations.

2.3 Conceptual Framework and Glossary

- Ancash Quechua (Waywash): A linguistic variety of Quechua spoken in the Ancash department and surrounding areas, characterized by the preservation of long vowels and a distinctive suffix system.
- Intelligent Agent: An autonomous software entity capable of perceiving its environment, making decisions, and communicating with other agents to achieve a common goal.
- RAG (Retrieval-Augmented Generation): An architectural pattern in which an LLM queries structured information stored in embeddings before generating a response.
 - Language Models (LLM): Deep neural networks trained on vast amounts of text that are capable of understanding and generating natural language.

III. Research Paradigm and Approach

This research is grounded in the Design Science Research (DSR) paradigm, as its primary purpose is not merely to describe or explain an educational phenomenon, but to design, develop, and validate an innovative technological artifact capable of solving a real-world problem: the need for adaptive and contextualized learning of Ancash Quechua using artificial intelligence.

Within this paradigm, scientific knowledge is generated through the systematic construction of technological solutions that are subsequently rigorously evaluated in real-world application contexts. Consequently, the research posits that the validity of knowledge stems not only from the observation or measurement of reality but also from the designed artifact's ability to effectively address a specific need.

In this study, the artifact is an Intelligent Tutor based on Multi-Agent Architecture and Large Language Models (LLMs). Its aim is to personalize the teaching-learning process of Ancash Quechua through specialized intelligent agents that interact in a coordinated manner to adapt content, assess student progress, and generate contextualized feedback. The Design Science paradigm is particularly relevant because it integrates principles from Artificial Intelligence, Multi-Agent Systems, Natural Language Processing, and Educational Technology, enabling the transformation of theoretical knowledge into a functional solution amenable to scientific evaluation.

Justification of the Paradigm

The choice of the Design Science Research paradigm is based on the following project characteristics:

- The primary objective is to build an intelligent tutor, constituting an innovative technological artifact.
- A multi-agent architecture is proposed, in which different agents collaborate to manage adaptive learning.
- Large Language Models (LLMs) are integrated to comprehend, generate, and contextualize responses in Ancash Quechua.
- The research entails a systematic evaluation of the artifact using objective indicators related to learning performance and user interaction.

- The results aim to yield both a technological contribution and a scientific contribution regarding the design of intelligent systems for indigenous languages.

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